Body Performance Analytics and Intelligent Classification System

A Comparative Study of Five Machine Learning Models

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Abstract

This report presents a comprehensive machine learning analysis of the Body Performance Dataset, comprising 13,393 records across 12 physiological and fitness-related attributes. Five algorithms were implemented and evaluated on two parallel tasks: multi-class classification of physical performance categories (A–D) and regression prediction of broad jump distance. The models studied are K-Nearest Neighbors, Linear Regression, Decision Tree, Support Vector Machine, and Neural Network. Each was assessed under 80:20, 70:30, and 50:50 train–test splits and 5-fold cross-validation. Classification accuracies ranged from ~59% to 72%; regression R^2 scores reached up to 0.81. A consistent accuracy ceiling near 71% was observed across all classifiers, attributed to class overlap, low-signal features, and label boundary ambiguity — all explained in detail below.

1. Introduction

Physical fitness classification is a practical application of machine learning in health informatics. The Body Performance Dataset records physiological measurements — age, gender, blood pressure, grip force, flexibility, sit-up endurance, and explosive power — mapped to a four-tier performance grade (A through D).

This project follows the complete ML pipeline: data understanding, preprocessing, exploratory data analysis, model training, cross-validation, and comparative evaluation. The objective is both to achieve competitive accuracy and to rigorously explain why a ~71% accuracy ceiling emerges uniformly across all five classifiers.

2. Dataset Overview and Exploratory Data Analysis

2.1 Dataset Summary

Property	Value
Total Rows	13,393
Columns	12
Numeric Features	10
Categorical Features	2 (gender, class)
Missing Values	None
Duplicate Rows	1 (removed)
Target Variable	class (A, B, C, D)
Class Balance	~25% each — balanced

2.2 Preprocessing Decisions

- Duplicate removal: 1 exact duplicate row removed → 13,392 rows.
- Validity violations: 2 rows systolic ≤ 40 mmHg and 6 rows diastolic ≤ 40 mmHg removed; 1 body fat > 70% removed; sit-and-bend capped to [−25, 80] cm. Final clean dataset: 13,385 rows.
- Type verification: all 10 numeric columns confirmed float64; gender and class confirmed object (categorical) — no corrections needed.
- Outlier treatment: height_cm (10 outliers, 0.07%) capped at IQR fences (likely measurement error); weight_kg and body_fat_% retained as clinically valid values.
- Encoding: gender → binary (F=0, M=1); class labels → integer (A=0, B=1, C=2, D=3).
- Feature scaling: StandardScaler applied for KNN, SVM, and Neural Network; raw values used for Decision Tree and Linear Regression.

2.3 Key EDA Insights

- Class balance: ~3,346–3,348 samples per class — perfectly balanced, no SMOTE or reweighting required.
- Strongest predictors: sit_and_bend_forward_cm ($r = -0.61$), sit_ups_counts ($r = -0.45$), broad_jump_cm ($r = -0.26$) with performance class.
- Low-signal features: diastolic and systolic BP have $r < 0.07$ with class, contributing minimal discriminative power.
- Age distribution: right-skewed (skewness = 0.60); older participants correlate with lower fitness classes.
- Gender dimorphism: height_cm is bimodal (male/female peaks); males score higher on grip force and broad jump on average.

3. Machine Learning Models

Model 1 — K-Nearest Neighbors (KNN)

KNN classifies each instance by majority vote among its k nearest neighbors in scaled feature space. StandardScaler was applied. Values of k from 1 to 99 were systematically evaluated across 80:20, 70:30, and 50:50 splits and 5-fold cross-validation.

Classification Results

Table 1a: KNN — k Tuning and Accuracy

k Value	Test Accuracy	Notes
k=3	55.8%	Overfitting — train accuracy higher than test
k=5	59.5%	Improved generalization
k=7	61.3%	Better bias–variance balance

k Value	Test Accuracy	Notes
k=9	61.2%	Accuracy continues rising with larger k
Optimized (scaled, k=22)	63.09%	Best k by CV on 80:20 split

Table 1b: KNN Classification Report (k=22, 80:20 split)

Class	Precision	Recall	F1-Score	Support
A (Excellent)	0.64	0.83	0.72	669
B (Good)	0.47	0.48	0.48	669
C (Average)	0.56	0.53	0.55	670
D (Poor)	0.92	0.66	0.77	669
Macro Avg	0.65	0.63	0.63	2,677

Regression Results — Predicting broad_jump_cm

Table 1c: KNN Regression Results

Split	R^2 Score	MAE (cm)
80:20	0.786	13.8 (MAE)
70:30	0.788	13.8 (MAE)
50:50	0.785	14.0 (MAE)

Accuracy Ceiling Analysis

KNN suffers from the curse of dimensionality across 11 features. Classes B and C overlap in feature space with no hard boundary, and all features receive equal weight including low-signal ones (blood pressure). Extensive k-tuning (k=1–99) was performed across all three splits and 5-fold CV. The optimal k was 22 (80:20) and 34 (CV), yielding a best accuracy of 63.09% and macro F1 of 0.63. Despite thorough tuning, no configuration exceeded 63.2%.

Model 2 — Linear Regression

Linear regression was applied in its native regression role (predicting broad_jump_cm) and adapted for classification by discretizing continuous outputs into performance bins.

Regression Results — Predicting broad_jump_cm (Primary Task)

Table 2a: Linear Regression — Regression Results (best split: 70:30, $R^2 = 0.8033$)

Split	MSE	RMSE (cm)	R^2 Score
80:20	324.95	18.03	0.7999
70:30	319.87	17.89	0.8033
50:50	319.90	17.89	0.7992
5-Fold CV avg	—	—	0.7872

Classification Results (Adapted)

Table 2b: Linear Regression — Adapted Classification Results

Split	Adapted Accuracy	Notes
80:20	~49%	Continuous output binned into class ranges
70:30	~51%	Marginal improvement with larger training set
50:50	~47%	Insufficient training data

Accuracy Ceiling Analysis

The 20% unexplained variance in broad_jump_cm reflects factors absent from the dataset: training history, technique, and motivation. The linear assumption is violated by non-linear fitness dynamics (e.g., diminishing returns with age). For classification, mapping a continuous output to ordinal class bins is a fundamental task mismatch that cannot be overcome by tuning.

Model 3 — Decision Tree

Decision trees recursively partition the feature space. Cost-complexity pruning (alpha) was optimised via 5-fold cross-validation. An unpruned tree (depth 28, 3,983 nodes) achieves 100% training accuracy but severely overfits.

Classification Results — Split Experiments

Table 3a: Decision Tree — Pruning and Split Experiments

Configuration	Train Acc	Test Acc	Depth	Nodes
Unpruned (alpha=0)	100.0%	64.94%	28	3,983
Underfit (depth=8)	—	63.18%	8	69
Optimal (alpha=0.000377)	—	71.07%	14	477
80:20 (optimal)	78.04%	72.15%	14	477
70:30	79.08%	70.58%	14	—
50:50	85.96%	67.06%	14	—
5-Fold CV avg	—	70.39%	—	—

Table 3b: Decision Tree Classification Report (optimal pruned model, 80:20 split)

Class	Precision	Recall	F1-Score	Support
A (Excellent)	0.71	0.85	0.77	837
B (Good)	0.58	0.58	0.58	837
C (Average)	0.67	0.62	0.65	838
D (Poor)	0.90	0.78	0.84	837
Overall Accuracy	—	—	0.71	3,349

Regression Results — Predicting broad_jump_cm

Table 3c: Decision Tree — Regression Results

Tree Depth	RMSE (cm)	R^2 Score	Notes
4 (underfit)	24.1	0.68	Too coarse
6 (optimal)	20.1	0.76	Best regression depth
12 (overfit)	22.8	0.72	Noisy splits hurt test

Accuracy Ceiling Analysis

Even with optimal pruning, axis-aligned decision boundaries cannot separate classes B and C whose feature ranges overlap substantially. The 50:50 split drops to 67%, revealing sensitivity to training set size. Deeper trees increase overfitting without improving test accuracy.

Model 4 — Support Vector Machine (SVM)

SVM maximises the margin between classes. Linear and RBF kernels were compared. Grid search covered $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$ with 3-fold cross-validation.

Classification Results — Kernel and Hyperparameter Comparison

Table 4a: SVM — Kernel and Hyperparameter Comparison

Kernel	C	γ	Test Accuracy	Notes
Linear (baseline)	—	—	63.13%	No tuning
Linear (scaled)	1	—	65.33%	After StandardScaler
RBF	1	0.1	68.4%	Moderate tuning
RBF	10	0.1	71.27%	Best configuration
RBF (3-fold CV)	10	0.1	70.8%	Consistent with test result

Table 4b: SVM Classification Report (RBF, C=10, γ=0.1, 80:20 split)

Class	Precision	Recall	F1-Score
A (Excellent)	0.73	0.84	0.78
B (Good)	0.58	0.57	0.57
C (Average)	0.64	0.60	0.62
D (Poor)	0.88	0.83	0.85
Weighted Avg	0.71	0.71	0.71

Regression Results — SVR on broad_jump_cm

Table 4c: SVR — Regression Results

Kernel	C	Epsilon	RMSE (cm)	R^2 Score
RBF	10	0.1	18.4	0.79
Linear	10	0.1	19.1	0.77
RBF	1	0.1	19.3	0.76

Accuracy Ceiling Analysis

C was swept from 0.01 to 100, polynomial kernels of degree 2–4 were tested, and gamma was varied across four values. No configuration exceeded 72.15%. The SVM kernel mapping cannot resolve a non-separable four-class manifold where classes B and C share overlapping physiological ranges — confirming the ceiling is data-intrinsic, not model-bound.

Model 5 — Neural Network (MLP)

A Multi-Layer Perceptron was implemented in TensorFlow/Keras with BatchNormalization, Dropout regularisation, ReduceLROnPlateau, and EarlyStopping (patience=20). Multiple architectures were explored and a data leakage issue discovered during validation is documented below. The reported results reflect the corrected, leak-free training pipeline.

Architecture Summary

Table 5a: Neural Network — Architecture Configurations

Architecture	Hidden Layers	Units	Activation	Regularisation
Shallow MLP (initial)	1	128	ReLU + Softmax	Dropout (0.3)
Deep MLP (final)	2	256 → 128	ReLU + Softmax	BatchNorm + Dropout (0.4, 0.35)
Optimised MLP	2	512 → 256 → 128 → 64	ReLU + Softmax	BatchNorm + Dropout + EarlyStopping

Classification Results

Table 5b: Neural Network — Classification Results

Split	Train Acc	Val Acc	Test Acc
80:20	~82%	~74%	74.76%
70:30	~80%	~74%	74.80%
50:50	~77%	~73%	73.33%
5-Fold CV avg	—	—	73.98% (±0.47%)

Regression Results — Predicting broad_jump_cm

Table 5c: Neural Network — Regression Results (R^2=0.802 across all splits, best regression model)

Split	RMSE (cm)	R^2 Score	Notes
80:20	17.77	0.802	Consistent across splits
70:30	17.77	0.802	Stable generalisation
50:50	17.79	0.802	No degradation with larger test set

Accuracy Ceiling Analysis

The initial deep MLP (512 units, 4 layers) produced 100% accuracy on all splits due to severe data leakage: the data pipeline inadvertently included the target encoding as a feature, combined with an embedded Random Forest preprocessing step that used the labels during transformation. Once corrected, the validated model (256→128→64 units, BatchNorm, Dropout, EarlyStopping with patience=20) achieved a consistent test accuracy of 74.76–74.80% across 80:20 and 70:30 splits, and 73.98% on 5-fold CV. The 73–75% range represents a genuine improvement over the previously reported ceiling, attributable to BatchNorm stabilisation and optimal early stopping. The remaining gap is explained by irreducible class B/C overlap and low-signal blood pressure features.

4. Comparative Analysis

4.1 Classification Performance — All Models

Table 6: Classification Summary — All Five Models

Model	Best Accuracy	Macro F1	Best Split	CV Accuracy
KNN (k=22)	63.09%	0.63	80:20	61.9%
Decision Tree (pruned)	72.15%	0.71	80:20	70.39%
SVM (RBF, C=10)	71.27%	0.71	80:20	70.8%
Neural Network (MLP)	74.80%	~0.74	70:30	73.98%
Linear Reg. (adapted)	~50%	~0.48	N/A	N/A

4.2 Regression Performance — All Models

Table 7: Regression Summary — All Five Models

Model	Best R^2	Best RMSE (cm)	Best Split
Neural Network	0.802	17.77	80:20 / 70:30
Linear Regression	0.8033	17.89	70:30
SVM (SVR)	0.79	18.4	80:20
Decision Tree	0.76	20.1	80:20
KNN Regression	0.788	13.8 (MAE)	70:30

4.3 The ~71% Accuracy Ceiling — Root Cause Analysis

All four dedicated classifiers converge to the same ceiling of ~70–72% regardless of model complexity or tuning depth. This convergence is informative: it points to a data-intrinsic limit rather than a modelling failure.

- Class B/C Overlap: F1 for class B (0.53–0.58) and C (0.42–0.65) is consistently the lowest across all models. These adjacent fitness tiers share overlapping physiological ranges with no hard boundary.
- Low-signal features: Blood pressure variables ($r < 0.07$) occupy model capacity without contributing discrimination. Feature ablation experiments show ≤1% accuracy gain from their removal.
- Label boundary ambiguity: Performance classes were discretised from continuous evaluation scores. Participants near class boundaries can legitimately belong to either adjacent tier.
- Extensive tuning performed: SVM swept C=0.01–100 and three kernels; Decision Tree optimised alpha via CV; KNN swept k=1–20; Neural Network tested depths 1–3, dropout 0.1–0.5, LR 0.0001–0.01. No configuration exceeded 72.15%.

5. Conclusions and Recommendations

- Best classifier: Neural Network (MLP, 74.80%) achieves the highest legitimate accuracy after correcting a data leakage bug that had falsely inflated results to 100%. Decision Tree (72.15%) is the best interpretable model; SVM (RBF, 71.27%) is preferred for deployment robustness.
- Best regression model: Neural Network and Linear Regression both achieve $R^2 \approx 0.802$ on broad_jump_cm prediction. Linear Regression is preferred for interpretability at equivalent performance; the NN shows no additional benefit for regression on this dataset.
- Lowest performers for classification: Linear Regression (~50%, task mismatch) and KNN (63.09%) are not recommended. KNN regression, however, performs comparably to other models ($R^2=0.788$).
- Future work: The ~71% ceiling is data-bound, not model-bound. Improvements should focus on feature engineering (composite fitness ratios, BMI) or collecting additional predictive attributes rather than increasing model complexity.

6. Academic Integrity Declaration

AI tools were used in an assistive capacity for report formatting and language editing only. All machine learning implementations, experimental results, and analytical interpretations are the original work of the team. Source code notebooks are submitted alongside this report for reproducibility verification.

References

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